Linux & Shell Lab 1: First Steps in the Terminal

Student worksheet | Name: _____ Group: _____ Date: _____

Welcome! In this lab you will learn to control a Linux computer by typing commands. No experience needed. Follow the parts in order, **type every command yourself** (do not copy and paste), and check the result after each step.

What you will learn

- What Linux, the shell and the terminal are
- How to move around the filesystem with `pwd`, `ls` and `cd`
- How to create, copy, move, rename and delete files and folders
- How to write and read simple text files
- How to find help on your own

Golden rules: Linux is case-sensitive. Press **Enter** to run a command. There is no Recycle Bin: deleted means gone.

Part 0: Meet the terminal

Word	Meaning	Think of it as
Linux	The operating system that manages files, programs and hardware	The engine of a car
Shell (usually `bash`)	A program that reads your commands and asks Linux to run them	The driver who understands your instructions
Terminal	The window where you type and see text	The dashboard
Command	One instruction, such as `ls`	A single order to the driver

Open a terminal. On Ubuntu press `Ctrl + Alt + T`. On Windows with WSL, open the Ubuntu app. You will see a line ending in `$`. This is the **prompt**: the shell is waiting for you.

```
student@pc:~$
```

How a command is written:

```
command  -option  argument
ls       -l       /home
```

- **command**: what to do
- **option** (starts with `-`): how to do it
- **argument**: what to do it on

Try it: run these four commands and write what each one printed.

```
whoami
date
echo Hello Linux
clear
```

Command	What it printed
`whoami`	
`date`	
`echo Hello Linux`	

Part 1: Where am I? Moving around

Linux stores everything in a tree of folders (called **directories**) that starts at one root, written `/`.

```
/                 <- root, the top of everything
|-- home/
|   `-- student/   <- your home directory (written ~)
|       |-- Documents/
|       `-- Downloads/
|-- etc/           <- system settings
`-- tmp/           <- temporary files
```

Command	What it does	Example
`pwd`	Prints the directory you are in now	`pwd`
`ls`	Lists what is inside a directory	`ls -l`
`cd`	Changes directory	`cd Documents`

Useful `ls` options: `-l` (detailed list), `-a` (show hidden files that start with a dot), `-h` (human-readable sizes). You can combine them: `ls -lah`.

Special paths:

- `~` is your home directory
- `.` is the current directory
- `..` is the parent directory (one level up)
- A path starting with `/` is **absolute**; otherwise it is **relative** to where you are

Try it, in this order:

```
pwd
ls
cd /
ls
cd /home
pwd
cd ~
pwd
cd Documents
cd ..
cd
pwd
```

Answer in your own words:

1. What does `cd` with no argument do? _____
2. What is the difference between `cd /tmp` and `cd tmp`? _____
3. Which folder are you in after `cd ../..` from `/home/student`? _____

Tip: `cd documents` fails if the folder is called `Documents`. Watch your capital letters.

Part 2: Create and organize files

You will now build your own workspace. Everything happens inside a practice folder, so nothing important can be damaged.

Command	What it does	Example
`mkdir`	Creates a directory	`mkdir lab1`
`touch`	Creates an empty file	`touch notes.txt`
`cp`	Copies a file or folder	`cp notes.txt backup.txt`
`mv`	Moves or renames	`mv backup.txt old.txt`
`rm`	Deletes a file (permanent)	`rm old.txt`
`rmdir`	Deletes an empty directory	`rmdir empty_dir`

Step by step. Type each line, and run `ls` after every step to see what changed.

```
cd ~
mkdir lab1
cd lab1
mkdir photos docs
touch docs/notes.txt docs/todo.txt
ls docs
cp docs/notes.txt docs/notes_backup.txt
mv docs/todo.txt docs/tasks.txt
ls docs
rm docs/notes_backup.txt
ls docs
```

What happened?

- `mkdir photos docs` created two directories in one command
- `cp` kept the original and made a second file
- `mv` did not copy anything: it **renamed** the file (moving to another folder works the same way)
- `rm` removed the file with no confirmation and no undo

Folders need an option. Try `rm photos`, read the error message, then run:

```
rmdir photos
```

To delete a folder that has content, use `rm -r foldername`. It removes everything inside, so be careful.

Safety warning: never run `rm -rf /` or `rm -rf *`. Run `pwd` before any `rm -r` to check where you are. A safe habit is `rm -i`, which asks for confirmation.

Try it: inside `lab1`, create a folder `project` with subfolders `src` and `data` in a single command (hint: `mkdir -p project/src project/data`), then check with `ls -R project`.

Part 3: Write and read text files

Most things in Linux are text files, so reading and writing them is a core skill.

Command	What it does	Example
`echo`	Prints text (or writes it to a file with `>`)	`echo "Hello" > hi.txt`
`cat`	Prints a whole file	`cat hi.txt`
`less`	Reads a long file page by page (quit with `q`)	`less /etc/passwd`

Command	What it does	Example
`head` / `tail`	First or last 10 lines	`head -n 3 file.txt`
`wc -l`	Counts lines	`wc -l file.txt`
`nano`	A simple text editor	`nano notes.txt`

Redirection sends output into a file instead of the screen:

- `>` writes to a file and **overwrites** it
- `>>` **appends** to the end of a file

```
cd ~/lab1/docs
echo "Linux is fun" > notes.txt
cat notes.txt
echo "Second line" >> notes.txt
echo "Third line" >> notes.txt
cat notes.txt
wc -l notes.txt
head -n 1 notes.txt
tail -n 1 notes.txt
```

Predict, then check: what will `cat notes.txt` show after you run `echo "Oops" > notes.txt`? Write your guess: _____. Now run it and compare.

Using nano. Open a file with `nano notes.txt`, type some text, then press `Ctrl + O` and Enter to save, and `Ctrl + X` to exit.

The pipe `|` sends the output of one command into the next one:

```
ls /etc | head -n 5
ls /etc | wc -l
```

Try it: create `about_me.txt` with three lines (your name, your city, your favorite subject) using `echo`, `>` and `>>`. Show it with `cat`, then count its lines with `wc -l`.

Part 4: Get help and work faster

Nobody memorizes every option. Learn to find answers yourself.

```
ls --help
man ls
```

In a `man` page, scroll with the arrow keys, search with `/word` then Enter, and quit with `q`.

Shortcut	Effect
`Tab`	Auto-completes file and command names (press twice to see options)
`Up` / `Down` arrows	Go through previous commands
`Ctrl + C`	Stops a running command
`Ctrl + L`	Clears the screen (same as `clear`)
`Ctrl + A` / `Ctrl + E`	Jump to start / end of the line

Type `history` to see your past commands, and `!!` to repeat the last one.

Try it:

1. Type `cd ~/lab1/do` and press `Tab`. What happened? _____
2. Use `man cp` to find the option that asks before overwriting a file. Option: _____

3. Run `sleep 30`, then stop it with `Ctrl + C`.
4. Use the Up arrow to bring back an earlier command and run it again.

Part 5: Mini challenge

Work alone or with a partner. Do not look at the earlier steps; use `--help`, `man` and your notes if you get stuck. Build this structure from scratch:

```
school/
|-- math/
|   `-- exercises.txt   (contains: "Exercise 1: 2+2")
|-- physics/
|   `-- formulas.txt    (contains: "E = mc^2")
`-- readme.txt          (contains 2 lines: your name, today's goal)
```

Tasks:

1. Go to your home directory and create the structure above
2. Copy `readme.txt` to `readme_backup.txt`, then rename the backup to `old_readme.txt`
3. Display the content of all three text files with `cat`
4. Show the full tree with `ls -R school`
5. Delete `old_readme.txt`, then run `ls school` to prove it is gone

When you finish, show your screen with `ls -R school` to your teacher. **Bonus:** add a folder `chemistry` and move `formulas.txt` into it without breaking anything.

Cheat sheet

Goal	Command
Where am I?	`pwd`
List files (detailed, hidden)	`ls -la`
Go into a folder / up one level / home	`cd name` / `cd ..` / `cd`
Make a folder (with parents)	`mkdir name` / `mkdir -p a/b/c`
Create an empty file	`touch file`
Copy a file	`cp source destination`
Move or rename	`mv source destination`
Delete a file / folder with content	`rm file` / `rm -r folder`
Show a file / page by page	`cat file` / `less file`
First / last lines	`head -n 5 file` / `tail -n 5 file`
Write / append text to a file	`echo "text" > file` / `echo "text" >> file`
Count lines	`wc -l file`
Edit a file	`nano file`
Get help	`command --help` / `man command`
Clear screen / stop a command	`clear` / `Ctrl + C`

Homework: My Linux Library

Goal: practice every command from today on your own, in about 45 minutes. Use the terminal only, not a graphical file manager.

Part A: Build the structure

1. In your home directory, create a folder `library` with three subfolders: `books`, `movies`, `music`
2. In each subfolder, create a text file (`books.txt`, `movies.txt`, `music.txt`) and write at least **3 favorite titles**, one per line, using `echo` with `>` and `>>`
3. Create a file `library/about.txt` containing your full name, your group, and today's date (hint: `date >> about.txt` appends the date)

Part B: Manipulate

4. Copy `books/books.txt` to `books/books_backup.txt`
5. Rename `movies/movies.txt` to `movies/films.txt`
6. Create a folder `library/archive` and move `books_backup.txt` into it
7. Use `head -n 2` and `tail -n 1` on `music.txt` and write what each command printed
8. Use `wc -l` to count the lines of each of your three title files

Part C: Explore and explain (short written answers)

9. Run `ls -la ~`. What are the files whose names start with a dot? Why do you think they are hidden?
10. Use `man ls` to find which option sorts files by modification time. Write the option and test it.
11. Explain in your own words the difference between `>` and `>>`, and between `rm` and `rmdir`.
12. Explore `/etc` with `ls /etc | head -n 10`. List three names you see and guess what each might be for.

Bonus (optional): find out what `cp -r` does and use it to copy the whole `library` folder to `library_copy`. Then explain why `cp library library_copy` alone would fail.

What to submit (one text file). As the final step, from your home directory, run:

```
ls -R library > submission.txt
history | tail -n 40 >> submission.txt
```

Then open `submission.txt` with `nano`, add your written answers for Part C at the bottom, and send it to your teacher by the deadline.

Deadline: _____

Reminders: the terminal is case-sensitive, use `Tab` to complete names, run `pwd` when you feel lost, and never use `rm -r` on a folder you did not create.